

SESSION

7

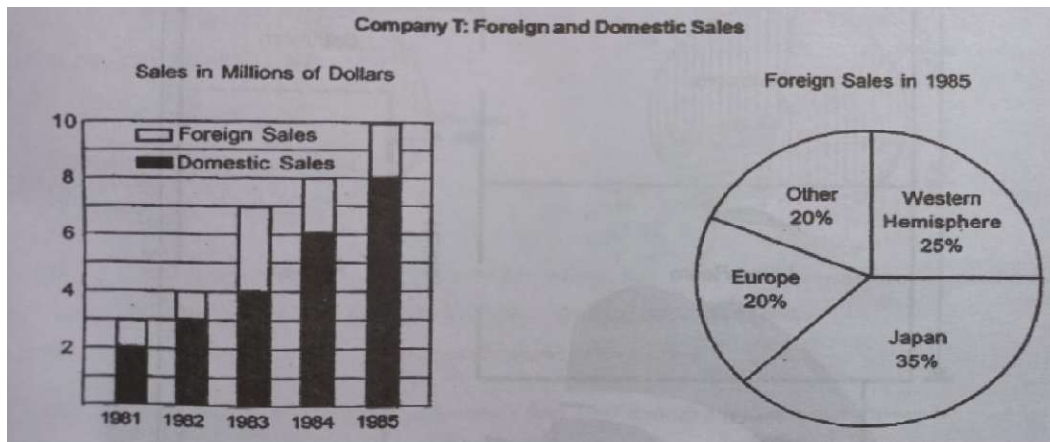


DI-LR

Learning Objectives

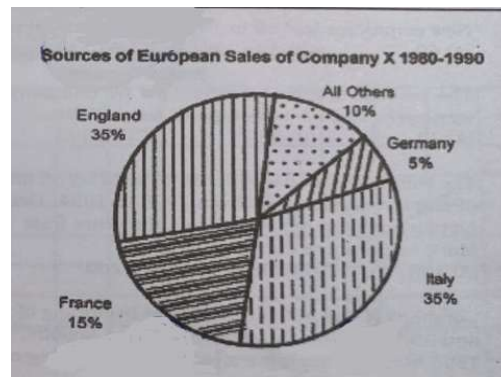
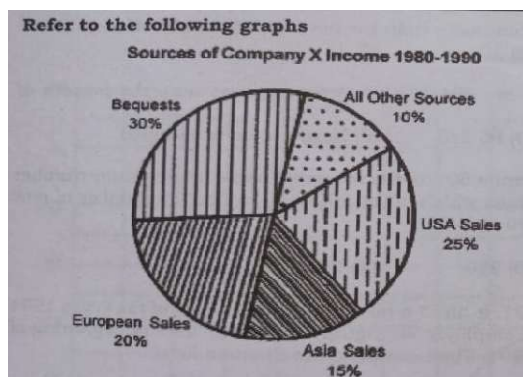
1. Data Interpretation
2. Blood Relations
3. Number Series

DIRECTIONS for Q. 1 to 7: For the following questions, five options are given. Choose the correct option.



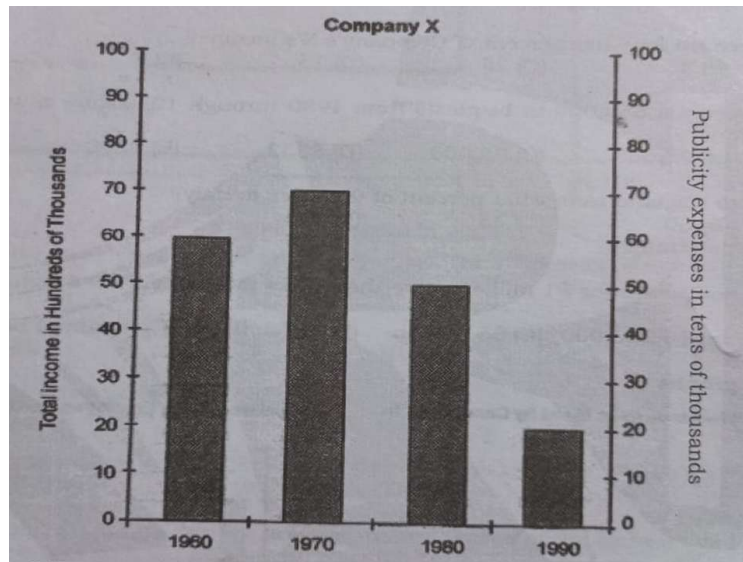
1. What was the total dollar value of foreign sales by company T in 1981 and 1982?
 (a) \$1,000,000 (b) \$2,000,000
 (c) \$3,000,000 (d) \$7,000,000
 (e) \$12,000,000
2. What was the difference in the dollar value of foreign sales by Company T between 1983 and 1985?
 (a) \$1,000,000 (b) \$2,000,000
 (c) \$3,000,000 (d) \$4,000,000
 (e) \$6,000,000
3. In 1984, foreign sales accounted for what percent of total sales by Company T?
 (a) 15% (b) 20%
 (c) 25% (d) 33.33%
 (e) 40%
4. In 1985, what was the dollar value of sales by Company T to Europe?
 (a) \$200,000 (b) \$400,000
 (c) \$1,200,000 (d) \$1,600,000
 (e) \$2,000,000
5. If sales to the Western Hemisphere in 1983 accounted for the same percent of foreign sales as Japan in 1985, what was the dollar value of sales to the Western Hemisphere in 1983?
 (a) \$1,050,000 (b) \$1,250,000
 (c) \$1,375,000 (d) \$1,425,000
 (e) \$1,555,000
6. Which of the following statements are definitely true about the sales of company T? (Mark one or more correct options)
 (a) Domestic sales increased by the same percentage from 1981 to 1982 and from 1983 to 1984.
 (b) Change in foreign sales from 1981 to 1982 was same as the change from 1984 to 1985.
 (c) Total sales changed by the same percentage from 1981 to 1982 and from 1983 to 1984.
 (d) All of the above statements are incorrect.
7. Sales to Germany contributed to 40% of the total European sales of company T in 1985. Determine the dollar value of the sales to Germany in 1985.
 _____ million dollars.

DIRECTIONS for Q. 8 to 12: For the following questions, five options are given. Choose the correct option.



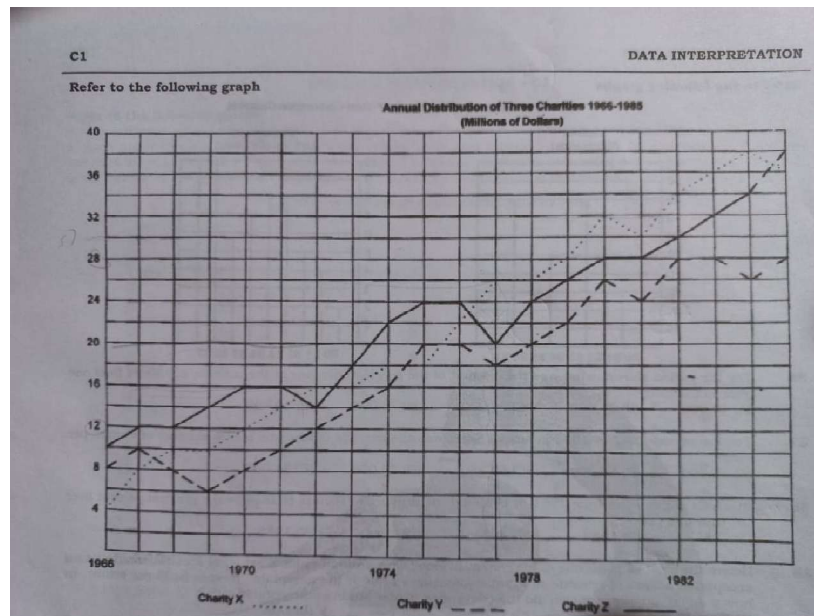
8. If total income from 1980 to 1990 were \$10 mn, how much of that income came from sales to Germany?
 (a) \$10 mn (b) \$51 mn
 (c) \$100,000 (d) \$10,000
 (e) \$1,000
9. Sales to Italy account for what percent of Company's X's income?
 (a) 87.5 (b) 35
 (c) 25 (d) 15
 (e) 7
10. If company X received \$50,000 in bequests from 1980 through 1990, how much money did it receive from sales to France?
 (a) \$16,666 (b) \$3,333
 (c) \$5,000 (d) \$333
 (e) \$250
11. The 1980 sales to England were what percent of the sales to Italy?
 (a) 100 (b) 50
 (c) 35 (d) 25
 (e) Can't be determined
12. If sales to Italy accounted for \$1 mn more than sales to France, how much income came from US sales?
 (a) \$200,000 (b) \$6,250,000
 (c) \$5 mn (d) \$25 mn
 (e) Can't be determined

DIRECTIONS for Q. 13 to 17: For the following questions, five options are given. Choose the correct option.



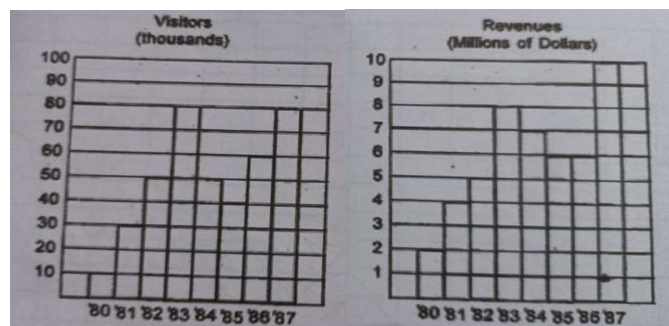
13. The total income of Company X from 1960-1990 inclusive was
 (a) \$20,000,000 (b) \$2,000,000
 (c) \$200,000 (d) \$20,000
 (e) \$200
14. In 1970, publicity expenses were approximately what percentage of total income of company X?
 (a) 10 (b) 13
 (c) 20 (d) 23
 (e) 40
15. If publicity expenses doubled in 1980, approximately what percentage of total income was spent on publicity expenses?
16. The amount of money spent by company X on television advertising in 1980 was
 (a) \$5,000,000 (b) \$700,000
 (c) \$500,000 (d) \$70,000
 (e) Can't be determined
17. Company X's publicity expenses in 1970 were what percent greater than those in 1990?
 (a) 500 (b) 250
 (c) 225 (d) 125
 (e) Can't be determined

DIRECTIONS for Q. 18 to 22: For the following questions, five options are given. Choose the correct option.



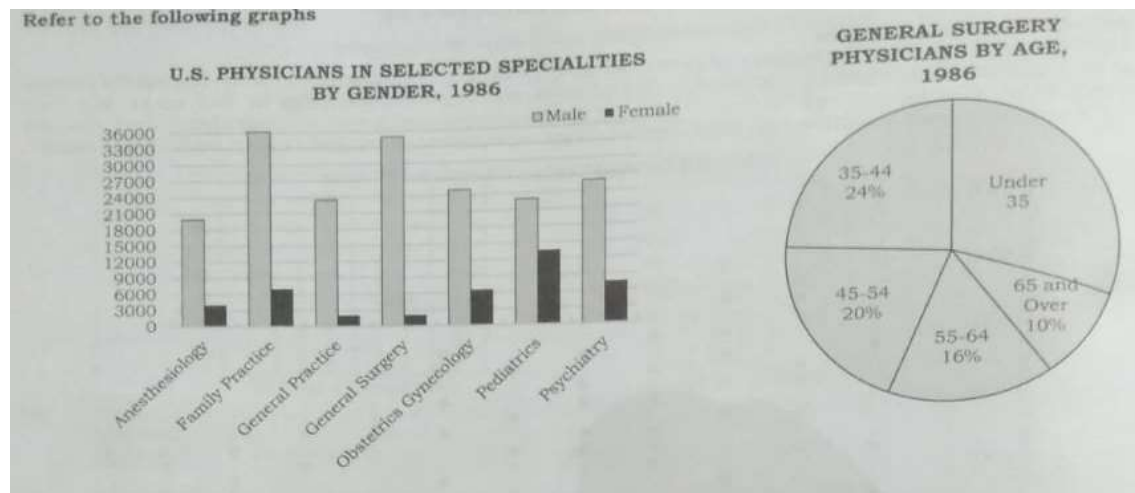
18. For how many years was the annual distribution of Charity X greater than those of both Charity Y and Charity Z?
(a) 4 (b) 9
(c) 11 (d) 14
(e) 15
19. From 1971 to 1982, by how many millions of dollars did the annual distribution by Charity Z increase?
(a) 8 (b) 12
(c) 14 (d) 16
(e) 18
20. From 1970 to 1981, by what percent did the annual distribution by Charity X increase?
(a) 100% (b) 150%
(c) 200% (d) 250%
(e) 300%
21. In 1977, what were the combined annual distributions of Charity X and Charity Z (in millions of dollars)?
(a) 38 (b) 44
(c) 46 (d) 52
(e) 64
22. Which of the following statements can be inferred from the information given?
I. Between 1969 and 1979, total annual disbursements by all three charities combined increased by more than 150 percent.
II. For each year after the first year shown, annual disbursement by Charity Y failed to increase in exactly four years.
III. For the years 1978 and 1979 combined, Charity X accounted for more than 1/3 of all money distributed by the three charities combined.
(a) Only I (b) Only II
(c) I and II only (d) I and III only
(e) I, II and III

DIRECTIONS for Q. 23 to 27: For the following questions, five options are given. Choose the correct option.



23. For the period shown, what was the amount of the greatest increase in the number of visitors from one year to the next?
 (a) 10,000 (b) 20,000
 (c) 30,000 (d) 70,000
 (e) 80,000
24. For the period 1982 to 1986 inclusive, what was the average (arithmetic mean) number of visitors per year?
 (a) 48,000 (b) 56,000
 (c) 59,250 (d) 75,000
 (e) 84,000
25. In which of the year from 1981 to 1985 did the number of visitors change by the greatest percent over the previous year?
 (a) 1981 (b) 1982
 (c) 1983 (d) 1984
 (e) 1985
26. Before the start of 1980, the management of Hotel Convention Centre K set what K considered to be an acceptable dollars-of-revenue to number-of-visitors ratio. If the acceptable ratio is \$135 per visitors, in how many of the year shown did the convention centre attain an acceptable ratio?
 (a) Five (b) Four
 (c) Three (d) Two
 (e) One
27. If, in 1984, 20% of all visitors accounted for 50% of the revenues, what was the average (arithmetic mean) amount of revenue derived from each of those visitors?
 (a) \$185 (b) \$215
 (c) \$280 (d) \$350
 (e) \$385

DIRECTIONS for Q. 28 to 30: For the following questions, five options are given. Choose the correct option.



28. If in 1986 all the family practice physicians represented 7.5% of all the physicians in the United States, approximately how many physicians were there total?
 (a) 300,000 (b) 360,000
 (c) 430,000 (d) 485,000
 (e) 570,000
29. If the number of female general surgeon physicians in the under-35 category represented 3.5 % of all the general surgeon physicians, approximately how many male general surgeon physicians were under 35 years?
 (a) 9,200 (b) 9,800
 (c) 10,750 (d) 11,260
30. In which specialty the female representation is maximum compared to the male representation in that specialty?
 (a) General Practice (b) Anesthesiology
 (c) Pediatrics (d) General Surgery

Answer the questions on the basis of the information given below.

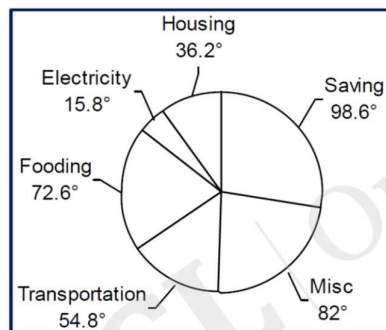
The table given below shows the percentage marks distribution of seven students in a monthly class test on various subjects. The figures given in the brackets for each subject represents the maximum marks for that particular subject.

	History (75)	Geography (75)	Maths (150)	Science (200)	English (100)	Hindi (50)
A	84	72	81	67	62	68
B	76	58	87	73	65	72
C	59	64	75	61	71	56
D	69	78	64	58	61	66
E	88	68	93	83	74	78
F	82	66	89	91	77	68
G	78	72	66	71	75	80

31. What is the total marks obtained by student "A" in History, Geography and Science together?
(a) 261 (b) 245
(c) 251 (d) 342
32. What is the percentage of total marks obtained by student "B" in all subjects together?
(a) 66.3% (b) 73.54%
(c) 76.84% (d) 72.75%
33. What is the (approx.) average marks obtained by students in Maths?
(a) 126 (b) 119
(c) 115 (d) 111
34. What is the (approx.) percentage of marks obtained by student "E" in History, Geography, Maths and Hindi together?
(a) 93.2% (b) 89.6%
(c) 81.4% (d) 84.4%
35. What is the difference between marks obtained by "C" and "G" in all subjects?
(a) 42.75 (b) 96.25
(c) 112.75 (d) 46

Answer the questions on the basis of the information given below.

The pie-chart given below shows the expenditure break-up of the total monthly income of Ms. Jyoti Mishra. Her total monthly income is Rs. 45,000.



36. The amount spent by Jyoti on "Fooding" and "Transportation" together constitute what percent of her monthly salary?
(a) 23.57% (b) 18.39%
(c) 35.39% (d) 45.21%
37. By what percentage, the amount spent on "Saving" is more than that of the amount spent on "Misc"?
(a) 20.24% (b) 36.71%
(c) 18.47% (d) 9.42%
38. The total amount spent on "Housing" and "Fooding" together is what percentage of amount spent on "Saving" and "Electricity" together?
(a) 76% (b) 81%
(c) 69% (d) 95.10%
39. If she cut down her "Misc" amount by 20% and invest this amount as a part of her "Saving" and rest of all the expenses remains the same, then the measure of the central angle formed by "Saving" is
(a) 108° (b) 101°
(c) 115° (d) 113°
40. What is combined amount spent on "fooding" and "housing"?
(a) Rs. 13,600 (b) Rs. 17,100
(c) Rs. 12,325 (d) Rs. 11,050



Class Practice Exercise 26

Data Interpretation - I

DIRECTIONS: Answer the questions on the basis of the information given below.

The table given below shows the percentage marks distribution of seven students in a monthly class test on various subjects. The figures given in the brackets for each subject represents the maximum marks for that particular subject.

	History (75)	Geography (75)	Maths (150)	Science (200)	English (100)	Hindi (50)
A	84	72	81	67	62	68
B	76	58	87	73	65	72
C	59	64	75	61	71	56
D	69	78	64	58	61	66
E	88	68	93	83	74	78
F	82	66	89	91	77	68
G	78	72	66	71	75	80

- What is the total marks obtained by student "A" in History, Geography and Science together?
(1) 261 (2) 245
(3) 251 (4) 342
- What is the percentage of total marks obtained by student "B" in all subjects together?
(1) 66.3% (2) 73.54%
(3) 76.84% (4) 72.75%
- What is the (approx.) average marks obtained by students in Maths?
(1) 126 (2) 119
(3) 115 (4) 111
- What is the (approx.) percentage of marks obtained by student "E" in History, Geography, Maths and Hindi together?
(1) 93.2% (2) 89.6%
(3) 81.4% (4) 84.4%
- What is the difference between marks obtained by "C" and "G" in all subjects?
(1) 42.75 (2) 96.25
(3) 112.75 (4) 46

DIRECTIONS: Answer the questions on the basis of the information given below.

Number of students appeared and qualified in an examination from five states over the year 1993 to 1998:

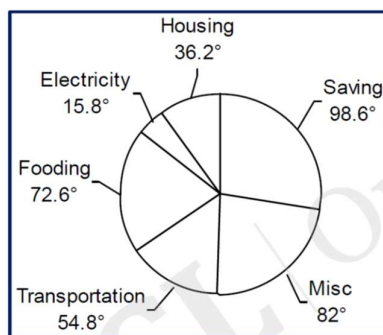
State	A		B		C		D		E	
	App.	Qual.	App.	Qual.	App.	Qual.	App.	Qual.	App.	Qual.
1993	950	230	880	200	1075	315	975	260	1120	325
1994	1090	320	960	250	1125	290	990	305	1208	308
1995	860	240	1100	330	950	275	1400	375	1375	380
1996	1200	350	1250	290	1240	325	1175	310	1060	265
1997	1150	300	1300	325	1220	340	1080	295	995	235
1998	980	280	1080	336	1140	305	1150	280	1075	290
TOTAL	6230	1720	6570	1731	6750	1850	6770	1825	6833	1803

- Approximately, what is the percentage of students qualified from states B and C together in the year 1996?
(1) 32% (2) 20%
(3) 25% (4) 30%

- The total number of students appeared from state A during the years 1996 to 1998 is approximately what percentage of the total number of students appeared from the same state during the years 1993 to 1995?
(1) 98% (2) 105%
(3) 85% (4) 115%
- The total number of students qualified from all the states in year 1998 is approximately what percentage of the number of students appeared from all the states in the same year?
(1) 25% (2) 27.5%
(3) 32% (4) 30%
- The number of students qualified from state D during the years from 1993 to 1995 is approximately what percentage of the students appeared from state D in these years?
(1) 27% (2) 32%
(3) 30% (4) 28%
- The percentage of students qualified is the highest for which state for all the six years together?
(1) A (2) B
(3) C (4) D

DIRECTIONS: Answer the questions on the basis of the information given below.

The pie-chart given below shows the expenditure break-up of the total monthly income of Ms. Jyoti Mishra. Her total monthly income is Rs. 45,000.



- The amount spent by Jyoti on "Fooding" and "Transportation" together constitute what percent of her monthly salary?
(1) 23.57% (2) 18.39%
(3) 35.39% (4) 45.21%
- By what percentage, the amount spent on "Saving" is more than that of the amount spent on "Misc"?
(1) 20.24% (2) 36.71%
(3) 18.47% (4) 9.42%
- The total amount spent on "Housing" and "Fooding" together is what percentage of amount spent on "Saving" and "Electricity" together?
(1) 76% (2) 81%
(3) 69% (4) 95.10%

14. If she cut down her "Misc" amount by 20% and invest this amount as a part of her "Saving" and rest of all the expenses remains the same, then the measure of the central angle formed by "Saving" is
 (1) 108° (2) 101°
 (3) 115° (4) 113°
15. What is combined amount spent on "fooding" and "housing"?
 (1) Rs. 13,600 (2) Rs. 17,100
 (3) Rs. 12,325 (4) Rs. 11,050

DIRECTIONS: Answer the questions on the basis of the information given below.

The bar graph shows the import and the export of a country over the years.

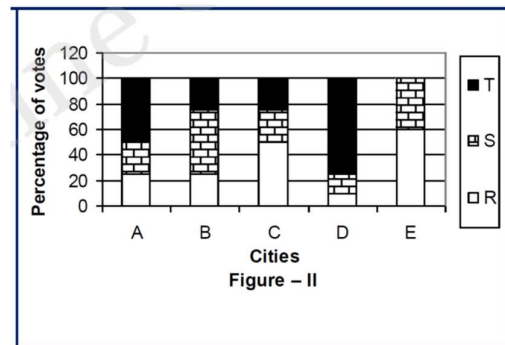
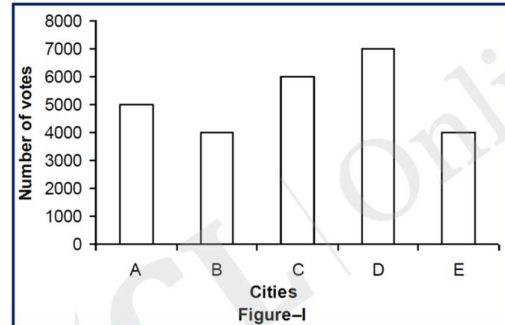


16. During which year is the percentage increase/decrease in imports over the previous year, lowest?
 (1) 1996 (2) 1997
 (3) 1994 (4) 1993
17. What is the percentage increase in exports in the year 1994 over the previous year?
 (1) 37.5% (2) 27.27%
 (3) 32.5% (4) 37.25%
18. What is the difference between the combined exports and imports during the years 1992, 1994 and 1997?
 (1) Rs. 400 crores (2) Rs. 350 crores
 (3) Rs. 250 crores (4) Rs. 450 crores
19. What is the annual growth rate in exports over the given period?
 (1) 5.55% (2) 33.33%
 (3) 4.76% (4) 25%
20. For which year, the ratio of imports to exports minimum?
 (1) 1991 (2) 1993
 (3) 1996 (4) 1997

DIRECTIONS: Answer the questions on the basis of the information given below.

There are two bar graphs given below. Figure (I) shows the total number of votes cast for the three candidates (R, S and T) in five different cities (A, B, C, D and E) in a statewide election. Figure (II) shows the distribution of votes among these three candidates in these five cities.

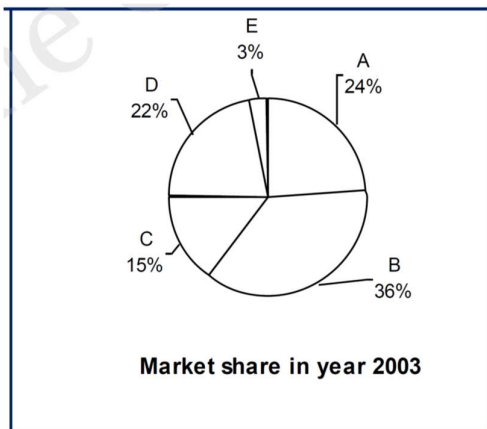
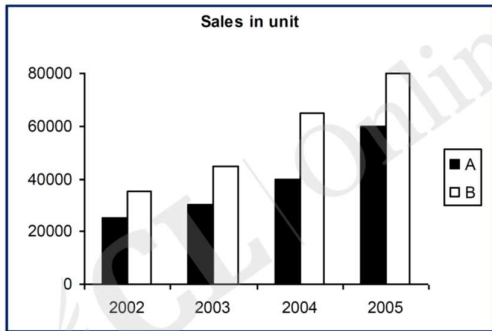
For questions 21 to 25



21. Which city cast the greatest number of votes for candidate R?
 (1) A (2) B
 (3) C (4) D
22. From how many cities did candidate S receive more votes than candidate T?
 (1) 2 (2) 1
 (3) 3 (4) 4
23. Which two cities cast the same number of votes for candidate R?
 (1) A and B (2) B and C
 (3) C and D (4) All cities were different
24. About how many votes did candidate T receive in total from five cities?
 (1) 4000-6000 (2) 6000-8000
 (3) 8000-10000 (4) 10000-12000
25. How many cities cast over two-thirds of their votes for one candidate?
 (1) 0 (2) 1
 (3) 2 (4) 3

DIRECTIONS: Answer the questions on the basis of the information given below.

The pie-chart given below shows the percentage distribution of market share of washing machine for five companies A, B, C, D and E in 2003. These are the only five companies in the market from year 2002 to 2005. The bar graph shows the sales in unit for two companies A and B over the given period.



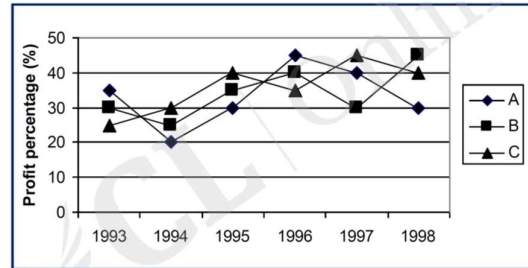
26. What was the sales of C in year 2003?
 (1) 48,000 units (2) 48,00 units
 (3) 12,000 units (4) 18,750 units
27. What was the average sales of B over the given period?
 (1) 45,000 units (2) 56,500 units
 (3) 37,500 units (4) None of these
28. What was the average annual growth rate of A over the period from 2002 to 2005?
 (1) 71.4% (2) 15%
 (3) 46.7% (4) 57.6%
29. What was the sales of E in year 2004?
 (1) 3,500 units (2) 6,250 units
 (3) 40,000 units (4) Can't be determined
30. What was the total volume of washing machine in year 2003?
 (1) 1,600 units (2) 8,000 units
 (3) 7,200 units (4) 125,000 units

DIRECTIONS: Answer the questions on the basis of the information given below.

The following graph shows the profit percentage earned by companies A, B and C over the given years.

Profit = Income – Expenditure

$$\text{Percentage profit} = \frac{\text{Profit} \times 100}{\text{Expenditure}}$$



31. If the income of company B in year 1994 was Rs. 26 lakhs, then what was the profit earned in that year?
 (1) Rs. 5.2 lakhs (2) Rs. 4.33 lakhs
 (3) Rs. 6.5 lakhs (4) Rs. 7.8 lakhs
32. If expenditure of company B in year 1995 is equal to the expenditure of company C in year 1998, then what will be the ratio of the income of company B in year 1995 to the income of company C in year 1998?
 (1) 8 : 9 (2) 27 : 28
 (3) 15 : 14 (4) 9 : 8
33. Income of company C in year 1993 and year 1994 were equal. What was the ratio of its expenditure in year 1993 to that of in year 1994?
 (1) 25 : 26 (2) 12 : 13
 (3) 13 : 12 (4) None of these
34. Income of company A in year 1996 is equal to its expenditure in year 1997. What was the ratio of its income in year 1996 to that in year 1997?
 (1) 0.4 (2) 0.6
 (3) 0.71 (4) 0.24
35. In which year is the percent increase/decrease in the profit percentage earned by company A minimum?
 (1) 1994 (2) 1995
 (3) 1996 (4) 1997

Class Practice Exercise 27

Data Interpretation – II

DIRECTIONS: Answer the questions on the basis of the information given below.

Each of the five friends – A, B, C, D and E – purchased a pair of shoes from a shoe store during Sale. E purchased Reebok shoes on 50% discount and spent same amount as D who purchased Nike shoes. The ratio of original price of shoes of brand Fila, Nike and Reebok is 4 : 5 : 6 respectively. A purchased Adidas shoes at 10% discount while B purchased Puma shoes at 20% discount. Original price of Fila shoes was Rs. 5,200, which was 4% more than that of Adidas and Rs. 1,150 more than that of Puma.

- What is the discounted price of Nike shoes?
(a) Rs. 3,900 (b) Rs. 4,050
(c) Rs. 5,200 (d) Rs. 4,500
- Who among the following purchased Fila shoes?
(a) D (b) A
(c) B (d) C
- If discounted price of Fila shoes is Rs. 3,640, then the discount percentage on Fila shoes is
(a) 30% (b) 40%
(c) 10% (d) 25%
- What was the discount percentage offered on Nike shoes?
(a) 50% (b) 40%
(c) 30% (d) 20%
- On which of the following brands maximum discount was offered?
(a) Adidas (b) Reebok
(c) Nike (d) Can't be determined

DIRECTIONS: Answer the following questions on the basis of the information given below.

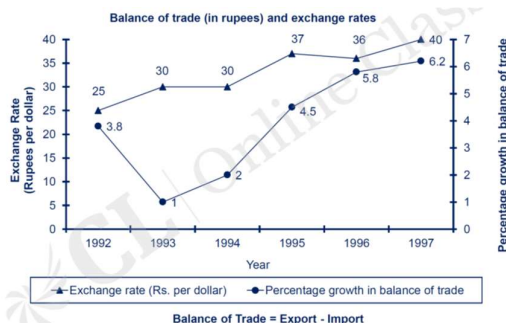
The following table compiles the annual figures related to Cocoa and Sugar production in selected states of Brazil for the year 2006-07.

States	Total Area	Percentage of Area under Cocoa production	Cocoa Production	Percentage of Area under Sugar production	Sugar Production	Population
	In Ten Thousand Hectares	%	Thousand Tons	%	Thousand Tons	Millions
Amapa	21	5	1.3	8	2.4	6
Brasilia	48	10	4.9	6	3.5	33
Mato Grosso	90	12	12.5	15	7.2	46
Maranhao	31	8	5.6	10	6.4	17
Parana	25	20	6.1	4	0.9	12
Rondonia	33	15	5.6	25	8	24
Sergipe	19	12	2.1	12	2.2	9
Tocantins	27	10	2.6	9	3.3	11
Goiás	34	6	7	13	15.8	10
Piaui	25	8	2.9	5	2.5	7
Amazonas	160	2	3.6	5	11.1	29
Santa Catarina	19	5	1.4	7	1.9	13
TOTAL	532		55.6		65.2	217

- How many of the mentioned states failed to cross the threshold yield of 0.1 ton of Cocoa per hectare of area under Cocoa production?
(a) 0 (b) 1
(c) 2 (d) 3

- If the domestic consumption of Cocoa in Brazil is 60% of its production and the rest of Cocoa production gets exported at a price of \$2000 per ton, what is the revenue earned by Brazil through Cocoa exports? [Assume these mentioned states account for 80% of Brazil's Cocoa production]
(a) \$42.8 Million (b) \$55.6 Million
(c) \$49.2 Million (d) \$64.8 Million
- International price of Cocoa and Sugar are \$2000 per ton and \$200 per ton respectively. Which state earned the highest revenue through export of Cocoa and Sugar, if all states exported their entire Cocoa and Sugar production at the given international price?
(a) Goiás (b) Parana
(c) Mato Grosso (d) Maranhao
- How many states with a population less than 15 million, produce more Cocoa than Sugar?
(a) 1 (b) 2
(c) 3 (d) 4
- How many of the given states have a total area (in hectare) per ton of Sugar Production more than 100 and a population of at least 13 million?
(a) 1 (b) 2
(c) 3 (d) 4

DIRECTIONS: Answer the following questions on the basis of the information given below.



- In which year is the percentage change in the growth rate of balance of trade (in rupees) minimum?
(a) 1993 (b) 1994
(c) 1995 (d) 1997
- Exports in 1995 were Rs. 37,000 crore. If the value of imports is same for the years 1994 and 1995, find the exports in 1994.
(a) Rs. 34,300 crore (b) Rs. 34,800 crore
(c) Rs. 35,750 crore (d) Can't be determined
- If the balance of trade in 1997 is Rs. 10,600 crore, then the balance of trade in 1996, expressed in million dollars is
(a) \$2,500 million (b) \$2,770 million
(c) \$3,200 million (d) \$3,500 million

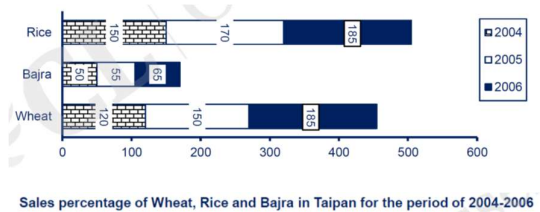
14. The value of rupee against the dollar is expected to decline by 10% in 1998. As a result, the value of \$1 would be
 (a) Rs. 44.44 (b) Rs. 44
 (c) Rs. 45.55 (d) Rs. 45
15. In 1997, the percentage decline in balance of trade, expressed in dollars, is approximately
 (a) 1% (b) 8.3%
 (c) 10% (d) 4.4%

DIRECTIONS: Answer the questions on the basis of the information given below.

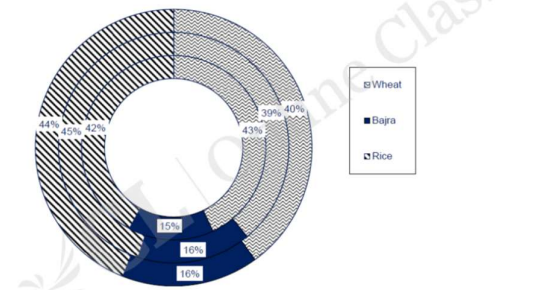
The production (in million tonnes) of wheat, rice and bajra for the country called Taipan for the years 2004 to 2006 are shown in the bar graph below. The country Taipan started production of these grains in 2004 only. The second figure shows the sales data for the mentioned three crops in Taipan for the years 2004 to 2006. The innermost ring represents the data for 2004, the middle ring represents the data for 2005 and the outer most ring represents the data for 2006. The total sales of all the crops taken together are 260, 335 and 375 million tonnes for the years 2004, 2005 and 2006 respectively.

Units available for sale in a year = Production in the year + Units unsold during last year

The production of wheat, Rice and Bajra in Taipan for the period 2004 to 2006 (in million tonnes)



Sales percentage of Wheat, Rice and Bajra in Taipan for the period of 2004-2006

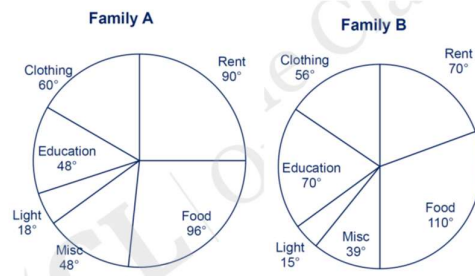


16. How much wheat remained unsold in 2004?
 (a) 8.2 Mn Tonne (b) 15.7 Mn Tonne
 (c) 4.3 Mn Tonne (d) 12.0 Mn Tonne
17. How much Bajra remained unsold at the end of the year 2006?
 (a) 30.2 Mn Tonne (b) 20.8 Mn Tonne
 (c) 17.4 Mn Tonne (d) 12.7 Mn Tonne
18. In 2005, sales as a percentage of production, is highest for which of the following crops?
 (a) Wheat (b) Rice
 (c) Both Rice and Bajra (d) Bajra

19. In 2005, sales as a percentage of units available for sale, is lowest for which of the following crops?
 (a) Wheat (b) Rice
 (c) Both Rice and Bajra (d) Both Wheat and Bajra
20. In 2007, if the sales of Rice increases by 10% as compared to 2006, what should be the percentage change in production of Rice in 2007 so that no Rice remained unsold at the end of 2007?
 (a) 10% increase (b) 1.9% decrease
 (c) 25.4% decrease (d) 45.2% decrease

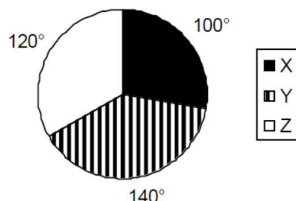
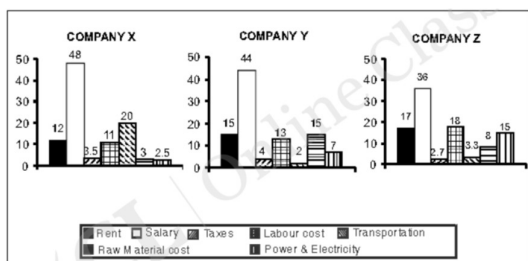
DIRECTIONS: Answer the following questions based on the given information.

The two pie diagrams given below provide the relative expenses of two families A and B.



21. If the total expenses of A and B are doubled, the ratio of expenditure on 'light' would be
 (a) 1 : 1 (b) 4 : 5
 (c) 2 : 3 (d) 5 : 4
22. If the expenses of B increase three-fold, the percentage of expense on education would be:
 (a) 19.55% (b) 20.44%
 (c) 20.54% nearly (d) 19.44%
23. The miscellaneous expenses of A as a percentage of those of B is
 (a) less than 75% (b) less than 80%
 (c) more than 80% (d) 80%
24. The item showing the least difference in expenditure between A and B is
 (a) food (b) light
 (c) clothing (d) miscellaneous
25. The percentages of expenses of A and B on food are in the respective ratio
 (a) 55 : 48 (b) 4 : 5
 (c) 5 : 4 (d) 48 : 55

DIRECTIONS: Answer the following questions based on the given information.

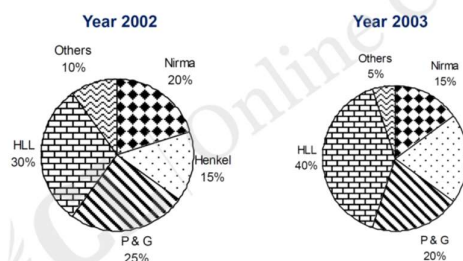


The given bar graphs show the percentage break-up of the annual expenditure of the three different companies viz. X India Ltd., Y India Ltd. and Z India Ltd. The pie-chart shows the expenditure break-up of each of the three companies. The total expenditure of the three companies is Rs.45,00,000.

26. Total salary expenditure of the three companies is what percentage of the total expenditure of the three companies?
 - (a) 37.47%
 - (b) 43.21%
 - (c) 41.92%
 - (d) 42.44%
27. If the average annual salary of a workers working in company X, Y and Z is Rs. 15,000, Rs. 11,000 and Rs.18,000 respectively, then what is the approximate number of workers working in the three companies?
 - (a) 511
 - (b) 140
 - (c) 222
 - (d) 129
28. By what percentage (approx.), the total expenditure of company Z on Power & Electricity is more or less than that spend by company Y on Power & Electricity?
 - (a) 68%
 - (b) 72%
 - (c) 80%
 - (d) 84%
29. By what percentage, the expense on 'Labour-Cost' is more or less than that on 'Rent' by the three companies?
 - (a) 10%
 - (b) 17%
 - (c) 5%
 - (d) 7.5%
30. What is the difference between the total expenditure on 'Taxes' to that of 'Raw Material' by the three companies?
 - (a) Rs. 7,29,060
 - (b) Rs.2,65,750
 - (c) Rs. 6,14,765
 - (d) Rs. 4,12,750

DIRECTIONS: Study the pie charts given below and answer the following questions.

The percentage market share on value basis of the companies in the detergent market for year 2002 and 2003 is given below.



The total size of the market has increased from Rs. 150 crore in year 2002 to Rs. 375 crore in year 2003.

31. What is the percentage increase in sales of the overall detergent market in year 2003 over 2002?
 - (a) 100%
 - (b) 150%
 - (c) 200%
 - (d) None of these
32. Which of the following had the minimum growth in sales in year 2003 over 2002?
 - (a) P & G
 - (b) HLL
 - (c) Henkel
 - (d) Others
33. What is the growth rate of HLL detergents in year 2003 over 2002?
 - (a) 10%
 - (b) 20%
 - (c) 30%
 - (d) None of these
34. What is the ratio of the sales of P & G for year 2002 to 2003?
 - (a) 1 : 2
 - (b) 2 : 1
 - (c) 3 : 1
 - (d) 1 : 2.5
35. If the sales value of all the companies in year 2002 would have been higher by 10%, then the detergent market would have been higher by how much?
 - (a) 15 crore
 - (b) 25 crore
 - (c) 30 crore
 - (d) 10 crore

Class Practice Exercise 28

Blood Relations

DIRECTIONS: For the following questions four options are given. Choose the correct option.

For Q. 1 to 3: D is the mother of S. S is the sister of T. T is the mother of R. R is the only son of J. J is the father of U. U is married to K.

1. How is J related to D?
 (1) Brother-in-law (2) Son-in-law
 (3) Grandson (4) Son
2. If R is father of X, then how K is related to X?
 (1) Uncle (2) Aunt
 (3) Grandson (4) Son
3. How is S related to R?
 (1) Sister-in-law (2) Mother-in-law
 (3) Grandmother (4) Aunt

For Q. 4 to 6: There are ten members A, B, C, D, E, F, G, H, I and J in the family. There are three generations of the family. There is also three married couple in the family. There is an equal number of male and female. A is the daughter – in – law of J.B's brother I have only one sister A. H and B are the married couple. H is the mother – in – law of C.G is the mother of B. D is the son in law of G. C is the sister of E. F is the only son of D. If B is the son in law of X.

4. Then how is X related to C?
 (1) Father (2) Grandmother
 (3) Grandfather (4) Can't be determined
5. If Y is the daughter of I. Then how is Y related to F?
 (1) Sister (2) Brother
 (3) Sister – in – law (4) Cousin
6. If Y is the father of I then how is D related to Y?
 (1) Son (2) Daughter
 (3) Son – in – law (4) Brother

For Q. 7 to Q. 9: Read the information carefully and answer the following questions.

M + N means M is the mother of N; M × N means M is the father of N; M \$ N means M is the wife of N.
 M % N means M is the sister of N; M ÷ N means M is the son of N; M # N means M is the husband of N.

7. What should come in place of question mark to establish that D is son-in-law of B in the expression 'A % B × C ? D × E' ?
 (1) ÷ (2) %
 (3) × (4) \$
8. Which of the following options is true if the expression 'A + B % J × L ÷ K' is definitely true?
 (1) L is daughter of B
 (2) K is son-in-law of A
 (3) A is grandmother of L
 (4) B is father of J

9. Which of the given expressions indicates that P is daughter of Q?

- (1) L % R % Q + T % P (2) L + R \$ Q × P % T
 (3) L % R \$ Q + T × P (4) B + L \$ Q ÷ P × T

For Q. 10 to Q. 12: Read the given information carefully and answer question that follow.

There are 7 family members P, Q, R, S, T, U and V standing in ground in which there are 2 married couples. P is sister of Q who is maternal grandson of T. Maternal grandfather of Q is standing 3m to the right of Q who is facing north. The father of S has 2 maternal grandchildren. V is facing north. V is standing 4m to the south of maternal grandson of U. S is 2m to the right of V. P is 1m south of S and 1m west of U. R is sister-in-law of V and standing 9m to the north of her mother. V is father of P. U is a Female.

10. Maternal Granddaughter of U is standing in which direction with respect to her husband?
 (1) South – East (2) South
 (3) North (4) South – West
11. What is a minimum distance between V and His father-in-Law?
 (1) 3 m (2) 2 m
 (3) 4 m (4) 6 m
12. What is the relation between S and U?
 (1) U is brother of S (2) S is sister of U
 (3) U is mother of S (4) S is son of U

For Q. 13 to Q. 30: Read the given information carefully and answer question that follow.

13. Pointing to a photograph of a boy Suresh said, "He is the son of the only son of my mother." How is Suresh related to that boy?
 (1) Brother (2) Uncle
 (3) Cousin (4) Father
14. If A + B means A is the mother of B; A - B means A is the brother B; A % B means A is the father of B and A × B means A is the sister of B, which of the following shows that P is the maternal uncle of Q?
 (1) Q - N + M × P (2) P + S × N - Q
 (3) P - M + N × Q (4) Q - S % P
15. If A is the brother of B; B is the sister of C; and C is the father of D, how D is related to A?
 (1) Brother (2) Sister
 (3) Nephew (4) Can't determine
16. If A + B means A is the brother of B; A - B means A is the sister of B and A × B means A is the father of B Which of the following means that C is the son of M?
 (1) M - N × C + F (2) F - C + N × M
 (3) N + M - F × C (4) M × N - C + F

17. Introducing a boy, a girl said, "He is the son of the daughter of the father of my uncle." How is the boy related to the girl?
 (1) Brother (2) Nephew
 (3) Uncle (4) Son-in-law
18. Pointing to a photograph Lata says, "He is the son of the only son of my grandfather." How is the man in the photograph related to Lata?
 (1) Brother (2) Uncle
 (3) Cousin (4) Data is inadequate
19. If $A + B$ means A is the brother of B; $A \times B$ means A is the son of B; and $A \% B$ means B is the daughter of A then which of the following means M is the maternal uncle of N?
 (1) $M + O \times N$ (2) $M \% O \times N + P$
 (3) $M + O \% N$ (4) None of these
20. If D is the brother of B, how B is related to C? To answer this question which of the statements is/are necessary?
 I. The son of D is the grandson of C.
 II. B is the sister of D.
 (1) Only I (2) Only II
 (3) Either I or II (4) I and II both are required
21. If $A + B$ means A is the father of B; $A - B$ means A is the brother B; $A \% B$ means A is the wife of B and $A \times B$ means A is the mother of B, which of the following shows that M is the maternal grandmother of T?
 (1) $M \times N \% S + T$ (2) $M \times N - S \% T$
 (3) $M \times S - N \% T$ (4) $M \times N \times S \% T$
22. Pointing to a photograph. Bajpai said, "He is the son of the only daughter of the father of my brother." How Bajpai is related to the man in the photograph?
 (1) Nephew (2) Brother
 (3) Father (4) Maternal Uncle
23. Deepak said to Nitin, "That boy playing with the football is the younger of the two brothers of the daughter of my father's wife." How is the boy playing football related to Deepak?
 (1) Son (2) Brother
 (3) Cousin (4) Brother-in-law
24. Pointing a photograph X said to his friend Y, "She is the only daughter of the father of my mother." How X is related to the person of photograph?
 (1) Daughter (2) Son
 (3) Nephew (4) Cannot be decided
25. Veena who is the sister-in-law of Ashok, is the daughter-in-law of Kalyani. Dheeraj is the father of Sudeep who is the only brother of Ashok. How Kalyani is related to Ashok?
 (1) Mother-in-law (2) Aunt
 (3) Wife (4) None of these
26. If $A + B$ means A is the sister of B; $A \times B$ means A is the wife of B, $A \% B$ means A is the father of B and $A - B$ means A is the brother of B. Which of the following means T is the daughter of P?
 (1) $P \times Q \% R + S - T$ (2) $P \times Q \% R - T + S$
 (3) $P \times Q \% R + T - S$ (4) $P \times Q \% R + S + T$
27. Pointing to a woman, Abhijit said, "Her granddaughter is the only daughter of my brother." How is the woman related to Abhijit?
 (1) Sister (2) Grandmother
 (3) Mother-in-law (4) Mother
28. Amit said - "This girl is the wife of the grandson of my mother". How is Amit related to the girl?
 (1) Brother (2) Grandfather
 (3) Husband (4) Father-in-law
29. A and B are children of D. Who is the father of A? To answer this question which of the statements I and II is necessary?
 I. C is the brother of A and the son of E.
 II. F is the mother B.
 (1) Only I (2) Only II
 (3) Either I or II (4) I and II both
30. Pointing towards a man, a woman said, "His mother is the only daughter of my mother." How is the woman related to the man?
 (1) Mother (2) Grandmother
 (3) Sister (4) Daughter
31. If $A + B$ means B is the brother of A; $A \times B$ means B is the husband of A; $A - B$ means A is the mother of B and $A \% B$ means A is the father of B, which of the following relations shows that Q is the grandmother of T?
 (1) $Q - P + R \% T$ (2) $P \times Q \% R - T$
 (3) $P \times Q \% R + T$ (4) $P + Q \% R - T$
32. $A3P$ means A is the mother of P; $A4P$ means A is the brother of P; $A9P$ means A is the husband of P; $A5P$ means A is the daughter of P. Which of the following means that K is the mother-in-law of M?
 (1) $M9N3K4J$ (2) $M9N5K3J$
 (3) $K5J9M3N$ (4) $K3J9N4M$
33. Pointing to a photograph Anjali said, "He is the son of the only son of my grandfather." How is the man in the photograph related to Anjali?
 (1) Brother (2) Uncle
 (3) Son (4) Data is inadequate
34. Pointing to a person, Deepak said, "His only brother is the father of my daughter's father". How is the person related to Deepak?
 (1) Father (2) Grandfather
 (3) Uncle (4) Brother-in-law
35. P is the mother of K; K is the sister of D; D is the father of J. How is P related to J?
 (1) Mother (2) Grandmother
 (3) Aunt (4) Data inadequate
36. If $P \$ Q$ means P is the father of Q; $P \# Q$ means P is the mother of Q and $P * Q$ means P is the sister of Q, then $N \# L \$ P * Q$ shows which of the relation of Q to N?
 (1) Grand son (2) Grand daughter
 (3) Nephew (4) Data is inadequate

37. If A \$ B means A is the brother of B; A @ B means A is the wife of B; A # B means A is the daughter of B and A * B means A is the father of B, which of the following indicates that U is the father-in-law of P?
- (1) P @ Q \$ T # U * W (2) P @ W \$ Q * T # U
(3) P @ Q \$ W * T # U (4) P @ Q \$ T # W * U
38. Introducing a man, a woman said, "He is the only son of the mother of my mother." How is the woman related to the man?
- (1) Mother (2) Sister
(3) Niece (4) Maternal aunt
39. Pointing to Gopi, Nalni says, "I am the daughter of the only son of his grandfather." How Nalni is related to Gopi?
- (1) Niece (2) Daughter
(3) Sister (4) Can't determine
40. A's son B is married with C whose sister D is married to E the brother of (2) How D is related to A?
- (1) Sister (2) Daughter's-in-law
(3) Sister-in-law (4) Cousin
41. Pointing to a lady a person said, "The son of her only brother is the brother of my wife." How is the lady related to the person?
- (1) Maternal aunt (2) Grandmother
(3) Sister of father-in-law (4) None of these
42. B5D means B is the father of D. B9D means B is the sister of D. B4D means B is the brother of D. B3D means B is the wife of D. Which of the following means F is the mother of K?
- (1) F3M5K (2) F5M3K
(3) F9M4N3K (4) F3M5N3K
43. A \$ B means A is the father of B; A # B means A is the sister of B; A * B means A is the daughter of B and A @ B means A is the brother of B. Which of the following indicates that M is the wife of Q?
- (1) Q \$ R # T @ M (2) Q \$ R @ T # M
(3) Q \$ R * T # M (4) Q \$ R @ T * M
44. If A \$ B means A is the brother of B; B * C means B is the son of C; C @ D means C is the wife of D and A # D means A is the son of D, how C is related to A?
- (1) Maternal grandmother (2) Maternal aunt
(3) Aunt (4) Mother
45. Pointing to a girl Sandeep said, "She is the daughter of the only sister of my father." How is sandeep related to the girl?
- (1) Uncle (2) Cousin
(3) Father (4) Grandfather
46. Pointing to a boy in the photograph Reena said, "He is the only son of the only child of my grandfather." How Reena is related to that boy?
- (1) Mother (2) Sister
(3) Aunt (4) Can't determine
47. A * B means A is the sister of B; A \$ B means B is the mother of A; A + B means A is the brother of B A = B means B is the father of A. Which of the following means M is the maternal uncle of N?
- (1) M = P + Q * N (2) N + P = Q * M
(3) N * P \$ Q * M (4) None of these
48. If M x N means M is the daughter of N; M + N means M is the father of N; M % N means M is the mother of N and M - N means M is the brother of N then P % Q + R - T x K indicates which relation of P to K?
- (1) Daughter-in-law (2) Sister-in-law
(3) Aunt (4) None of these
49. If P + Q means P is the brother of Q; P x Q means P is the father of Q and P - Q means P is the sister of Q, which of the following relations shows that I is the niece of K?
- (1) K + Y + Z - I (2) K + Y x I - Z
(3) Z - I x Y + K (4) K x Y + I - Z
50. Pointing towards a girl, Abhisek says, "This girl is the daughter of only a child of my father." What is the relation of Abhisek's wife to that girl?
- (1) Daughter (2) Mother
(3) Aunt (4) Sister
51. If A \$ B means B is the father of A; A # B means B is the mother of A; A * B means B is the sister of A and A @ B means B is the husband of A, which of the following indicates that N is the grandmother of P?
- (1) P * Q # M \$ N (2) P @ Q \$ M # N
(3) P # Q \$ N * M (4) None of these
52. Anupam said to a lady sitting in a car, "The only daughter of the brother of my wife is the sister-in-law of the brother of your sister." How the husband of the lady is related to Anupam?
- (1) Maternal uncle (2) Uncle
(3) Father (4) Son-in-law
53. If P + Q means P is the brother of Q; P x Q means P is the wife of Q and P % Q means P is the daughter of Q then which of the following means D is the uncle of A?
- (1) A % B x C + D (2) A x B + C % D
(3) A + C % B x D (4) None of these
54. A is the son of C; C and Q are sisters; Z is the mother of Q and P is the son of Z. Which of the following statements is true?
- (1) P and A are cousins
(2) P is the maternal uncle of A
(3) Q is the maternal grandfather of A
(4) C and P are sisters
55. M is the father of N who is the son of V. In order to know the relation of M to P, which of the statement/statements is/are necessary?
- I. P is the brother of V.
II. The daughter of N is the granddaughter of V.
- (1) Only I (2) Only II
(3) Either I or II (4) I and II both
56. If X + Y means X is the daughter of Y; X - Y means X is the brother of Y; X % Y means X is the father of Y and X x Y means X is the sister of Y. Which of the following means I is the niece of J?
- (1) J - N % C x I (2) I x C - N % J
(3) J + M x C % I (4) I x C + N - J

DIRECTIONS: Read the following information carefully and answer the questions that follow.

A + B means A is the mother of B. A – B means A is the sister of B. A * B means A is the father of B. A ? B means A is the brother of B.

57. Which of the following means Q is the grandfather of P?

- (1) P + N * M * Q (2) Q * N * M + P
(3) Q ? M ? N * P (4) None of these

58. Which of the following means that N is the maternal uncle of M?

- (1) N ? P – L + E – M (2) N – Y + A ? M
(3) M – Y * P – N (4) N ? C + F * M

DIRECTIONS: Read the following information carefully and answer the questions that follow.

M % N means M is the son of N. M @ N means M is the sister of N. M \$ N means M is the father of N.

59. Which of the following shows the relation that C is the granddaughter of E?

- (1) C % B \$ F \$ E (2) B \$ F \$ E % C
(3) C @ B % F % E (4) E % B \$ F \$ C

60. Which of the following shows the relation that S is the father of Q?

- (1) S @ P \$ Q (2) Q @ P % S
(3) Q \$ S @ P (4) None of these

Class Practice Exercise 29

Blood Relations II

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. Pointing to a woman, a man said, "Her father is the only son of my father". How is the man related to the woman?
(1) Brother (2) Son
(3) Grandfather (4) Father
2. Pointing to a boy, a lady says that his mother is the only sister of her mother. How is the boy related to that lady?
(1) Nephew (2) Uncle
(3) Cousin (4) Brother
3. If Akshay is the brother of the son of Sunil's son, then how is Akshay related to Sunil?
(1) Nephew (2) Brother
(3) Cousin (4) Grandson
4. If "P + Q" means "P is the brother of Q" "P × Q" means "P is the wife of Q" and "P ÷ Q" means "P is the daughter of Q", then which of the following represents "A is the son of D"?
(1) $A \div C \times B + D$ (2) $A + C \div B \times D$
(3) $A \times B + C \div D$ (4) $A \div B \times C + D$
5. If "P + Q" means "P is the father of Q" "P × Q" means "P is the sister of Q" and "P ÷ Q" means "P is the brother of Q", then which of the following represents "N is the nephew of M"?
(1) $M \times A \div B \times N$ (2) $M \times A + N \div B$
(3) $M \div N + B$ (4) $N + B \div M$
6. "A + B" means "A is the son of B". "A - B" means "A is the wife of B". "A ÷ B" means "A is the mother of B". "A × B" means "A is the brother of B", and "A = B" means "A is the sister of B".
What does P + Q - R mean?
(1) R is the mother of P (2) P is the son of R
(3) P is the uncle of R (4) None of these
7. "A + B" means "A is the son of B". "A - B" means "A is the wife of B". "A ÷ B" means "A is the mother of B". "A × B" means "A is the brother of B", and "A = B" means "A is the sister of B".
What does P × Q ÷ R mean?
(1) P is the father of R
(2) P is the nephew of R
(3) P is the maternal uncle of R
(4) None of these
8. Introducing a woman, a man said, "Her mother's husband's sister is my aunt." How is the man related to that woman?
(1) Nephew (2) Cousin
(3) Brother (4) Either brother or cousin
9. Pointing to a photograph a lady tells Manav, "I am the only daughter of this lady and her son is your maternal uncle." How is the speaker related to Manav's father?
(1) Sister (2) Wife
(3) Daughter (4) Mother

10. A is the brother of K, K is the brother of Q and R is the sister of Q. How is R related to A?
(1) Brother (2) Sister
(3) Cousin (4) Nephew
11. Introducing Harsh, Ritu said, "He is the only son of my mother's mother." How is Ritu related to Harsh?
(1) Sister (2) Nephew
(3) Niece (4) Either (2) or (3)
12. If "A + B" means "A is the son of B", "A - B" means "A is the daughter of B" and "A × B" means "A is the husband of B", then A + B × Q - S means
(1) "A is grandson of S" (2) "A is the son of Q"
(3) "B is son-in-law of S" (4) All of these
13. Pointing to a girl, Amit said, "This is the wife of the grandson of the lady who had me as her only son". How is Amit related to the girl?
(1) Daughter (2) Daughter-in-law
(3) Father (4) Father-in-law
14. Ramesh told Ashish, "Yesterday, I defeated the only brother, of the daughter of my maternal grandmother." Whom did Ramesh defeat?
(1) Maternal uncle (2) Father-in-law
(3) Sister (4) Grandfather
15. Pointing towards photograph of a person, Alka said, "He is the only son of the father of my sister's brother." How is that person related to Alka?
(1) Uncle (2) Mother
(3) Father (4) Brother

DIRECTIONS: Answer the following question based on the given information.

A family consists of five members: P, Q, R, S and T where T has two sons, an unmarried daughter and a daughter-in-law. P is the brother-in-law of above-mentioned daughter-in-law. Q's sister is not happy with Q's wife. But P and his father support Q's wife S.

16. Who is the daughter of T?
(1) P (2) Q
(3) R (4) S
17. What is the relation of P with S?
(1) Brother (2) Brother-in-law
(3) Sister-in-law (4) Sister
18. How is T related to Q?
(1) Father (2) Brother
(3) Father-in-law (4) Can't be determined
19. Who is the wife of Q?
(1) P (2) R
(3) S (4) Cannot be determined
20. How is P related to T?
(1) Father (2) Brother
(3) Sister (4) None of these

21. Rajiv is one of the brothers in law of Sunita's husband Prakash. Pinky is daughter of Sunita. How is Pinky's brother Ritesh related to Rajiv?

- (1) Nephew (2) Son
(3) Niece (4) Cousin

22. If $A + B$ means A is the brother of B; $A \times B$ means A is the father of B; $A \div B$ means A is the sister of B; then which of the following would mean x is the uncle of y?

- (1) $x \div y \times z$ (2) $x \times y + z$
(3) $x + z \times y$ (4) None of these

23. Pointing to a man, a lady says that his father is the third son of her grandmother. How is the lady related to that man?

- (1) Cousin (2) Sister
(3) Brother (4) Either sister or cousin

DIRECTIONS: Answer the following question based on the given information.

A family consists six members U, V, W, X, Y and Z. There are two married couples in the family.

i. X has two unmarried children.

ii. W is the grandmother of Y who is the sister of V.

iii. V loves his father Z.

24. If U has only a son, then how is U related to X?

- (1) Father (2) Daughter
(3) Daughter-in-law (4) Father-in-law

25. How many females are there in the family?

- (1) 2 (2) 3
(3) 4 (4) 3 or 4

26. How is Y related to Z?

- (1) Daughter (2) Son
(3) Father (4) None of these

27. How is X related to Z?

- (1) Husband (2) Daughter
(3) Wife (4) Father

Class Practice Exercise 30

Direction Sense I

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. One morning Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to his right then in which direction was Udai facing?
(1) East (2) West
(3) North (4) South
2. Y is in the East of X which is in the North of Z. If P is in the South of Z, then in which direction of Y, is P?
(1) North (2) South
(3) South-East (4) None of these
3. If South-East becomes North, North-East becomes West and so on. What will West become?
(1) North-East (2) North-West
(3) South-East (4) South-West
4. A man walks 5 km toward South and then turns to the right. After walking 3 km he turns to the left and walks 5 km. Now in which direction is he from the starting place?
(1) West (2) South
(3) North-East (4) South-West
5. Rahul put his timepiece on the table in such a way that at 6:00 am hour hand points to North. In which direction the minute hand will point at 9:15 pm?
(1) South-East (2) South
(3) North (4) West
6. Rasik walked 20 m towards North. Then he turned right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Finally, he turns left and walks 15 m. In which direction and how many metres is he from the starting position?
(1) 15 m West (2) 30 m East
(3) 30 m West (4) 45 m East
7. Two cars start from the opposite places of a main road, 150 km apart. First car runs for 25 km and takes a right turn and then runs 15 km. It then turns left and then runs for another 25 km and then takes the direction back to reach the main road. In the mean time, due to minor break down the other car has run only 35 km along the main road. What would be the distance between two cars at this point?
(1) 65 km (2) 75 km
(3) 80 km (4) 85 km
8. Starting from the point X, Jayant walked 15 m towards West. He turned left and walked 20 m. He then turned left and walked 15 m. After this he turned to his right and walked 12 m. How far and in which directions is now Jayant from X?
(1) 32 m, South (2) 47 m, East
(3) 42 m, North (4) 27 m, South
9. One evening before sunset Rekha and Hema were talking to each other face to face. If Hema's shadow was exactly to the right of Hema, which direction was Rekha facing?
(1) North (2) South
(3) East (4) Data is inadequate
10. A boy rode his bicycle Northward, then turned left and rode 1 km and again turned left and rode 2 km. He found himself 1 km West of his starting point. How far did he ride northward initially?
(1) 1 km (2) 2 km
(3) 3 km (4) 5 km
11. K is 40 m South-West of L. If M is 40 m South-East of L, then M is in which direction of K?
(1) East (2) West
(3) North-East (4) South
12. A man walks 2 km towards North. Then he turns to East and walks 10 km. After this he turns to North and walks 3 km. Again, he turns towards East and walks 2 km. How far is he from the starting point?
(1) 10 km (2) 13 km
(3) 15 km (4) None of these
13. The length and breadth of a room are 8 m and 6 m respectively. A cat runs along all the four walls and finally along a diagonal order to catch a rat. How much total distance is covered by the cat?
(1) 10 (2) 14
(3) 38 (4) 48
14. One morning Sujata started to walk towards the Sun. After covering some

Class Practice Exercise 31

Number Series I

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. 2, 1, $1/2$, $1/4$, ..(?)..
 (1) $(1/3)$ (2) $(1/8)$
 (3) $(2/8)$ (4) $(1/16)$
2. 7, 10, 8, 11, 9, 12, ..(?)..
 (1) 7 (2) 10
 (3) 12 (4) 13
3. 36, 34, 30, 28, 24, ..(?)..
 (1) 20 (2) 22
 (3) 23 (4) 26
4. 22, 21, 23, 22, 24, 23, ..(?)..
 (1) 22 (2) 24
 (3) 25 (4) 26
5. 53, 53, 40, 40, 27, 27, ..(?)..
 (1) 12 (2) 14
 (3) 27 (4) 53
6. 21, 9, 21, 11, 21, 13, 21, ..(?)..
 (1) 14 (2) 15
 (3) 21 (4) 23
7. 58, 52, 46, 40, 34, ..(?)..
 (1) 26 (2) 28
 (3) 30 (4) 32
8. 3, 4, 7, 8, 11, 12, ..(?)..
 (1) 7 (2) 10
 (3) 14 (4) 15
9. 8, 22, 8, 28, 8, ..(?)..
 (1) 9 (2) 29
 (3) 32 (4) 34
10. 31, 29, 24, 22, 17, ..(?)..
 (1) 15 (2) 14
 (3) 13 (4) 12
11. 1.5, 2.3, 3.1, 3.9, ..(?)..
 (1) 4.2 (2) 4.4
 (3) 4.7 (4) 5.1
12. 14, 28, 20, 40, 32, 64, ..(?)..
 (1) 52 (2) 56
 (3) 96 (4) 128
13. 2, 4, 6, 8, 10, ..(?)..
 (1) 11 (2) 12
 (3) 13 (4) 14
14. 201, 202, 204, 207, ..(?)..
 (1) 205 (2) 208
 (3) 210 (4) 211
15. 544, 509, 474, 439, ..(?)..
 (1) 404 (2) 414
 (3) 420 (4) 445

DIRECTIONS: In the following questions, identify the number which completes the SERIES.

16. 28 25 5 21 18 5 14 ..(?).. ..(?)..
 (1) 11 5 (2) 10 7
 (3) 11 8 (4) 5 10
17. 8 11 21 15 18 21 22 ..(?).. ..(?)..
 (1) 25 18 (2) 25 21
 (3) 25 29 (4) 24 21
18. 9 16 23 30 37 44 51 ..(?).. ..(?)..
 (1) 59 66 (2) 56 62
 (3) 58 66 (4) 58 65
19. 2 8 14 20 26 32 38 ..(?).. ..(?)..
 (1) 2 46 (2) 44 50
 (3) 42 48 (4) 40 42
20. 9 11 33 13 15 33 17 ..(?).. ..(?)..
 (1) 19 33 (2) 33 35
 (3) 33 19 (4) 15 33
21. 4 9 16 25 36 ..(?)..
 (1) 49 (2) 64
 (3) 100 (4) 121
22. 17 17 34 20 20 31 23 ..(?).. ..(?)..
 (1) 26 23 (2) 34 20
 (3) 23 33 (4) 23 28
23. 6 20 8 14 10 8 12 ..(?).. ..(?)..
 (1) 14 10 (2) 2 18
 (3) 4 12 (4) 2 14
24. 21 25 18 29 33 18 ..(?).. ..(?)..
 (1) 43 18 (2) 41 44
 (3) 37 18 (4) 37 41
25. 75 65 85 55 45 85 35 ..(?).. ..(?)..
 (1) 25 15 (2) 25 85
 (3) 35 25 (4) 85 35
26. 11 16 21 26 31 36 41 ..(?).. ..(?)..
 (1) 47 52 (2) 46 52
 (3) 45 49 (4) 46 51
27. 3 8 13 18 23 28 33 ..(?).. ..(?)..
 (1) 39 44 (2) 38 44
 (3) 38 43 (4) 37 42
28. 84 78 72 66 60 54 48 ..(?).. ..(?)..
 (1) 44 34 (2) 42 36
 (3) 42 32 (4) 40 34
29. 20 20 17 17 14 14 11 ..(?).. ..(?)..
 (1) 8 8 (2) 11 11
 (3) 11 14 (4) 11 8
30. 61 57 50 61 43 36 61 ..(?).. ..(?)..
 (1) 29 61 (2) 27 20
 (3) 31 61 (4) 29 22

31. 9 12 11 14 13 16 15 ..(?).. ..(?)..
 (1) 14 13 (2) 18 21
 (3) 14 17 (4) 18 17
32. 4 8 22 12 16 22 20 24 ..(?).. ..(?)..
 (1) 28 32 (2) 28 22
 (3) 22 28 (4) 32 36
33. 40 40 31 31 22 22 13 ..(?).. ..(?)..
 (1) 13 4 (2) 13 5
 (3) 4 13 (4) 9 4
34. 1 10 7 20 13 30 19 ..(?).. ..(?)..
 (1) 26 40 (2) 29 36
 (3) 40 25 (4) 25 31
35. 10 20 25 35 40 50 55 ..(?).. ..(?)..
 (1) 70 65 (2) 60 70
 (3) 60 75 (4) 65 70
36. 42 40 38 35 33 31 28 ..(?).. ..(?)..
 (1) 25 22 (2) 26 23
 (3) 26 24 (4) 25 23
37. 6 10 14 18 22 26 30 ..(?).. ..(?)..
 (1) 36 40 (2) 33 37
 (3) 38 42 (4) 34 38
38. 8 12 9 13 10 14 11 ..(?).. ..(?)..
 (1) 14 11 (2) 15 12
 (3) 8 15 (4) 15 19
39. 36 31 29 24 22 17 15 ..(?).. ..(?)..
 (1) 13 11 (2) 10 5
 (3) 13 8 (4) 10 8
40. 3 5 35 10 12 35 17 ..(?).. ..(?)..
 (1) 22 35 (2) 35 19
 (3) 19 35 (4) 19 24

DIRECTIONS: Look carefully for the pattern, and then choose which pair of numbers comes next.

41. 13 29 15 26 17 23 19 ..(?).. ..(?)..
 (1) 21 23 (2) 20 21
 (3) 20 17 (4) 25 27
42. 14 14 26 26 38 38 50 ..(?).. ..(?)..
 (1) 60 72 (2) 50 62
 (3) 50 72 (4) 62 62
43. 44 41 38 35 32 29 26 ..(?).. ..(?)..
 (1) 24 21 (2) 22 19
 (3) 23 19 (4) 23 20
44. 34 30 26 22 18 14 10 ..(?).. ..(?)..
 (1) 8 6 (2) 6 4
 (3) 14 18 (4) 6 2
45. 32 31 32 29 32 27 32 ..(?).. ..(?)..
 (1) 25 32 (2) 31 32
 (3) 29 32 (4) 25 30

DIRECTIONS: Look carefully for the pattern, and then choose which pair of numbers comes next.

46. 7 9 66 12 14 66 17 ..(?).. ..(?)..
 (1) 19 66 (2) 66 19
 (3) 19 22 (4) 20 66
47. 3 8 10 15 17 22 24 ..(?).. ..(?)..
 (1) 26 28 (2) 29 34
 (3) 29 31 (4) 26 31
48. 4 7 26 10 13 20 16 ..(?).. ..(?)..
 (1) 14 4 (2) 14 17
 (3) 18 14 (4) 19 14
49. 32 29 26 23 20 17 14 ..(?).. ..(?)..
 (1) 11 8 (2) 12 8
 (3) 11 7 (4) 32 29
50. 16 26 56 36 46 68 56 ..(?).. ..(?)..
 (1) 80 66 (2) 64 82
 (3) 66 80 (4) 78 68
51. 2 44 4 41 6 38 8 ..(?).. ..(?)..
 (1) 10 12 (2) 35 32
 (3) 34 9 (4) 35 10
52. 17 32 19 29 21 26 23 ..(?).. ..(?)..
 (1) 25 25 (2) 20 22
 (3) 23 25 (4) 25 22
53. 17 14 14 11 11 8 8 ..(?).. ..(?)..
 (1) 8 5 (2) 5 2
 (3) 8 2 (4) 5 5
54. 10 34 12 31 14 28 16 ..(?).. ..(?)..
 (1) 25 18 (2) 30 13
 (3) 19 26 (4) 18 20
55. 11 14 14 17 17 20 20 ..(?).. ..(?)..
 (1) 23 23 (2) 23 26
 (3) 21 24 (4) 24 24
56. F2, ..(?).., D8, C16, B32.
 (1) A16 (2) G4
 (3) E4 (4) E3
57. 664, 332, 340, 170, ..(?).. , 89, ...
 (1) 85 (2) 97
 (3) 109 (4) 178
58. V, VIII, XI, XIV, __, XX, ...
 (1) IX (2) XXIII
 (3) XV (4) XVII
59. 70, 71, 76, ..(?).. , 81, 86, 70, 91, ...
 (1) 70 (2) 71
 (3) 80 (4) 96
60. 8, 43, 11, 41, ..(?).. , 39, 17, ...
 (1) 8 (2) 14
 (3) 43 (4) 44
61. VI, 10, V, 11, __, 12, III, ...
 (1) II (2) IV
 (3) IX (4) 14

62. $\frac{1}{9}, \frac{1}{3}, 1, \dots(?)\dots, 9, \dots$

(1) $\frac{2}{3}$

(3) 6

(2) 3

(4) 27

63. 83, 73, 93, 63, $\dots(?)\dots$, 93, 43, $\dots$

(1) 33

(3) 73

(2) 53

(4) 93

64. 15, $\dots(?)\dots$, 27, 27, 39, 39, $\dots$

(1) 51

(3) 23

(2) 39

(4) 15

65. 72, 76, 73, 77, 74, $\dots(?)\dots$, 75, $\dots$

(1) 70

(3) 75

(2) 71

(4) 78

Class Practice Exercise 32

Number Series II

DIRECTIONS: In each of the following questions, a number series is given with one of the terms missing. Choose the correct alternative that will continue the same pattern and replace the question mark (?) in the given series.

- | | |
|---|--|
| <p>1. 21, 23, 27, 33, ?
 (1) 37 (2) 41
 (3) 43 (4) 49</p> <p>2. 68, 67, 64, 59, ?
 (1) 52 (2) 53
 (3) 54 (4) 56</p> <p>3. 0, 2, 6, 8, 16, 30, 54, ? (MA T 2011)
 (1) 74 (2) 88
 (3) 100 (4) 110</p> <p>4. 4, ?, 144, 400, 900, 1764 (SSC 2012)
 (1) 25 (2) 36
 (3) 49 (4) 100</p> <p>5. 2, 10, 30, 68, ? (SSC 2012)
 (1) 125 (2) 130
 (3) 128 (4) 135</p> <p>6. 6, 11, 21, 36, 56, ?
 (1) 42 (2) 51
 (3) 81 (4) 91</p> <p>7. 8, 14, 26, 44, 68, ?
 (1) 71 (2) 78
 (3) 84 (4) 98</p> <p>8. 3, 9, 6, 36, 30, ? (SSC 2012)
 (1) 900 (2) 800
 (3) 950 (4) 400</p> <p>9. 36, 34, 30, 28, 24, ? (SSC 2013)
 (1) 23 (2) 26
 (3) 20 (4) 22</p> <p>10. 97, 90, 76, 55, ?
 (1) 28 (2) 27
 (3) 26 (4) 25</p> <p>11. 1, 6, 15, ?, 45, 66, 91 (SSC 2011)
 (1) 25 (2) 26
 (3) 27 (4) 28</p> <p>12. 5, 11, 23, 47, 95, ?
 (1) 190 (2) 191
 (3) 161 (4) 169</p> <p>13. 7, 12, 19, 28, 39, ?
 (1) 52 (2) 50
 (3) 51 (4) 48</p> <p>14. 2, 6, 12, ?, 30 (SSC 2011)
 (1) 18 (2) 24
 (3) 20 (4) 26</p> | <p>15. 24, 6, 18, 9, 36, 9, 24, ? (UB Clerks 2011)
 (1) 24 (2) 12
 (3) 8 (4) 6</p> <p>16. 2, 1, 4, 3, 6, 5, 8, ?
 (1) 9 (2) 10
 (3) 7 (4) 8</p> <p>17. 1, 4, 27, 256, ?
 (1) 625 (2) 3125
 (3) 3025 (4) 1225</p> <p>18. 285, 253, 221, 189, ?
 (1) 122 (2) 153
 (3) 157 (4) 151</p> <p>19. 8, 3, 11, 14, 25, ?
 (1) 50 (2) 39
 (3) 29 (4) 11</p> <p>20. 3, 8, 27, 112, ?, 3396
 (1) 565 (2) 452
 (3) 560 (4) 1530</p> <p>21. 2, 9, 28, 65, ? (CDS 2012)
 (1) 121 (2) 195
 (3) 126 (4) 103</p> <p>22. 31, 13, 45, 54, ?, 63
 (1) 36 (2) 54
 (3) 61 (4) 58</p> <p>23. 8, 27, 64, ?
 (1) 216 (2) 224
 (3) 64 (4) 125</p> <p>24. 6, 9, 12, 15, 15, 18, ? (SSC 2013)
 (1) 21 (2) 20
 (3) 19 (4) 22</p> <p>25. 107, 97, 82, 62, ?
 (1) 52 (2) 42
 (3) 47 (4) 37</p> <p>26. 30, 68, 130, 222, ?, 520, 738 (SSC 2012)
 (1) 420 (2) 350
 (3) 250 (4) 280</p> <p>27. 2, 2, 29, 38, 47, ? (SSC CGL 2014)
 (1) 59 (2) 56
 (3) 52 (4) 58</p> <p>28. 17, 43, 81, 131, ?
 (1) 375 (2) 468
 (3) 300 (4) 193</p> <p>29. (3)5, 7, 10.5, 14, ? (SSC 2013)
 (1) 15.5 (2) 16.5
 (3) 18.5 (4) 17.5</p> |
|---|--|

30. 3, 128, 6, 64, 9, ?, 12, 16, 15, 8 (IB 2012)
 (1) 32 (2) 12
 (3) 108 (4) 72
31. 8, 16, 28, 44, ? (SSC 2014)
 (1) 62 (2) 64
 (3) 66 (4) 60
32. 2, 8, 18, 32, 50, ? (SSC 2013)
 (1) 64 (2) 72
 (3) 70 (4) 68
33. 1, 3, 7, 15, 31, ? (SSC 2014)
 (1) 37 (2) 36
 (3) 73 (4) 63
34. 0, 3, 8, 15, 24, ?, 48 (SSC 2014)
 (1) 41 (2) 29
 (3) 37 (4) 35
35. 4, 11, 30, 67, 128, ? (SSC 2013)
 (1) 219 (2) 228
 (3) 231 (4) 237
36. 5, 6, 9, 14, 21, ? (SSC 2013)
 (1) 28 (2) 30
 (3) 31 (4) 29
37. 5, 16, 51, 158, ? (SSC CGL 2013)
 (1) 1452 (2) 483
 (3) 481 (4) 1454
38. 11, 29, 55, ?, 131 (SSC 2013)
 (1) 110 (2) 81
 (3) 89 (4) 78
39. 198, 194, 185, 169, ? (SSC 2013)
 (1) 92 (2) 136
 (3) 144 (4) 112
40. 0, 1, 1, 2, 3, 5, 8, 13, 21, ? (CLAT 2013)
 (1) 34 (2) 35
 (3) 33 (4) 36
41. 17, 36, 74, 150, ?, 606 (CLAT 2013)
 (1) 250 (2) 303
 (3) 300 (4) 302
42. Find the missing number in the series
 4, 18,, 100, 180, 294
 1. 32 (2) 36
 (3) 48 (4) 40
43. 10, 14, 28, 32, 64, 68, 132 (CDS 2011)
 (1) 28 (2) 32
 (3) 64 (4) 132

44. 1788, 892, 444, 220, 112, 52, 24
 (1) 52 (2) 112
 (3) 220 (4) 444
45. 17, 36, 53, 83, 92 (SSC 2012)
 (1) 92 (2) 53
 (3) 68 (4) 83
46. 20, 40, 200, 400, 2000, 4000, 8000
 (1) 200 (2) 2000
 (3) 8000 (4) 4000

DIRECTIONS: In each of the following questions a logical series of numbers is given. Find the wrong term.

47. 2 3 10 38 172
 (1) 192 (2) 10
 (3) 38 (4) 253
48. 8 19 52 151 447
 (1) 19 (2) 8
 (3) 52 (4) 447
49. 4 2 3 7.5 26.75
 (1) 26.75 (2) 7.5
 (3) 3 (4) 2
50. 2 3 10 40 172 885 5346
 (1) 3 (2) 885
 (3) 40 (4) 172
51. 15 18 20 29 45 70 106
 (1) 18 (2) 20
 (3) 29 (4) 45
52. 11 14 20 29 40 56
 (1) 40 (2) 4
 (3) 10 (4) 19
53. 49 50 46 55 38 64 28
 (1) 49 (2) 46
 (3) 55 (4) 38
54. 6 15 41 179 839
 (1) 15 (2) 6
 (3) 179 (4) 41
55. 4 6 12 30 90 31(2)50 1260
 (1) 6 (2) 90
 (3) 30 (4) 315
56. 35 19 11 7 5 (4)5 (3)5
 (1) (4)5 (2) 5
 (3) 11 (4) 19

Class Practice Exercise 33

Number and Alphabet Series I

DIRECTIONS: In the following question, complete the given series.

1. 20, 17, 34, 31, 62, 59, ?
(1) 95 (2) 118
(3) 110 (4) 128
2. 0, 7, 26, 63, 124, ?
(1) 210 (2) 215
(3) 211 (4) 224
3. 1, 2, 6, 15, 31, 56, 92, ?
(1) 150 (2) 128
(3) 141 (4) 149
4. 4, 4, 12, 16, 36, 36, 108, ?
(1) 64 (2) 66
(3) 82 (4) 86
5. 8, 9, 11, 15, 16, 18, ?
(1) 21 (2) 22
(3) 23 (4) 24
6. 0, 4, 18, 48, 100, ?
(1) 200 (2) 180
(3) 160 (4) 210
7. 4, 5, 8, 17, 44, ?
(1) 102 (2) 104
(3) 125 (4) 110
8. 6, 14, 26, 42, 62, 86, ?
(1) 114 (2) 115
(3) 116 (4) 118
9. 1, 1, 2, 4, 3, 9, 4, 16, 5, 25, 6, ?
(1) 35 (2) 42
(3) 49 (4) 36
10. 5, 10, 15, 25, 40, 65, 105, 170, ?
(1) 290 (2) 275
(3) 205 (4) 180

DIRECTIONS: In the letter series given below, some letters are missing which are given in that order as one of the four alternatives under it. Find the correct alternative in each case.

11. b -- b - a b c - b - a
(1) baaab (2) cacac
(3) cbaac (4) baacb
12. a b -- b a a - b -- a a b
(1) aaaaa (2) aacaa
(3) cccaa (4) ccccc
13. a - b a b - b - a b -- b a b
(1) ccbbb (2) babab
(3) bccca (4) aabca
14. a -- a - b a b - a - b
(1) bbbbb (2) babab
(3) bbaab (4) aabaa

15. In the letter series given below, some letters are missing which are given in that order as one of the four alternatives under it. Find the correct alternative in each case. Find the missing alphabet:

- s, p, N, k, l, ?
(1) f (2) e
(3) g (4) D

16. In the letter series given below, some letters are missing which are given in that order as one of the four alternatives under it. Find the correct alternative in each case.

- P S V Y : T V X Z : : G J M P : ?
(1) JMOQ (2) JOMQ
(3) KMOQ (4) KMQO

17. In the following question a sequence of groups of letters and numbers is given with one term missing. Choose the missing term out of the given alternates.

- P2Z, R4X, T8U, V16Q, ?
(1) X32L (2) Y32L
(3) X32M (4) Y64L

18. In the following question a sequence of groups of letters and numbers is given with one term missing. Choose the missing term out of the given alternates.

- 12G4, 15J8, 21M16, 24P32, ?
(1) 29R32 (2) 30S64
(3) 29S64 (4) 30R64

19. In the following question a sequence of groups of letters and numbers is given with one term missing. Choose the missing term out of the given alternates.

- T2C, R7D, P14F, N23I, ?
(1) L36M (2) L34M
(3) L290 (4) M36K

20. In the following question a sequence of groups of letters and numbers is given with one term missing. Choose the missing term out of the given alternates.

- A1, E3, I7, O15, ?
(1) U32 (2) U31
(3) U23 (4) P29

21. Find the missing term in the series.

- PRT, ____, BDF, HJL, NPR
(1) UWY (2) VYA
(3) VXZ (4) UXW

22. Find the missing term in the series.

- VWU, ZAY, ____, KLJ, RSQ
(1) CDB (2) DEC
(3) EFD (4) GHF

23. Find the missing term in the series.

- VER, RIV, ____, FUH, ZAN
(1) LMN (2) MNA
(3) MOC (4) LOB

24. Find the wrong number in the series.
7, 13, 27, 53, 106, 213, 427
(1) 13 (2) 106
(3) 213 (4) 427

25. 3, 4, 0, 9, -8, 18, -18
(1) 0 (2) -8
(3) 18 (4) 4

26. 2, 4, 10, 20, 50, 90
(1) 20 (2) 10
(3) 4 (4) 90

DIRECTIONS: In each of the following questions a number series is given. Below this series, a number is given followed by (i), (ii), (iii), (iv) and (v). After understanding the pattern in the series, you have to form another number series in the same order, starting with the given number.

27. 2 3 10 15 26 35
50 (i) (ii) (iii) (iv) (v)
Which of the following numbers will come in the place of (iv)?
(1) 122 (2) 100
(3) 120 (4) 143

28. 8 8 16 48 192 960
15 (i) (ii) (iii) (iv) (v)
Which of the following numbers will come in the place of (iii)?
(1) 30 (2) 90
(3) 60 (4) 45

29. 255 252 84 81 27 24
390 (i) (ii) (iii) (iv) (v)
Which of the following numbers will come in place of (iv)?
(1) 126 (2) 39
(3) 129 (4) 42

30. 176, 190, ____, 252, 292
(1) 230 (2) 222
(3) 210 (4) 218